Governed Multimodal AI Project Design Report

Section 1 - Governance Controls Summary

Governance position statement: Governance controls are largely mature, with every dataset classified, stewarded and permissioned on least-privilege terms, but the report de-identification checkpoint lacks a transformation version and run ID, leaving lineage short of the controlled-production standard (GOV-LIN-03, CFG-DP-01).

Data Classification and Access Permissioning Summary

Dataset	Sensitivity Level	Care Context	Permitted Use	Owner/Steward	Retrieve Roles	Transform Roles	Link Roles	Evidence ID(s)
DICOM Metadata Extract	**Restricted Clinical Data** - direct identifiers present at source; de-identification required before AI use	Radiology operations (direct care / source-system operations)	Approved source processing and de-identification only; must remain in the restricted processing zone until de-identification is complete	Owner: Director, Radiology Informatics Steward: Imaging Data Steward Policy: Clinical Data Use Standard; AI Governance Policy (quarterly review)	Imaging Data Engineer ML Engineer: denied (source-zone RBAC)	Imaging Data Engineer	None - linking occurs only after de-identification	GOV-CLS-01; GOV-STW-01; GOV-ACC-01; GOV-ACC-02
Radiology Report Text	**Restricted Clinical Data** - patient and encounter identifiers present before redaction	Radiology operations (direct care / source-system operations)	Approved redaction pipeline only; source text may not be exposed to model-development roles before de-identification	Owner: Chief Medical Information Office Steward: Clinical Text Data Steward Policy: Clinical Data Use Standard; Privacy & De-identification Policy (quarterly review)	Clinical Text Data Engineer ML Engineer: denied (source-zone RBAC)	Clinical Text Data Engineer	None - ML Engineer receives redacted text only	GOV-CLS-02; GOV-STW-02; GOV-ACC-03; GOV-ACC-04
Structured Patient Attributes	**Confidential Derived Health Data** - no direct identifiers, but linked health information and residual quasi-identifiers remain	Clinical informatics / AI evaluation	Structured evaluation context inside the approved project boundary; every retained attribute must pass minimum-necessary review	Owner: Enterprise Data Office Steward: EHR Data Steward Policy: Data Minimization Standard; AI Governance Policy (quarterly review)	Clinical Informatics Analyst	Clinical Informatics Analyst	Clinical Informatics Analyst (validates linkage semantics)	GOV-CLS-03; GOV-STW-03; GOV-ACC-05; PRV-RISK-03
Linked De-identified Analysis Dataset	**Confidential Derived Health Data** - pseudonymous study key; residual re-identification risk remains	AI development and testing	Model evaluation and governance review; linkage permitted only inside the approved project workspace	Owner: AI Governance Committee Steward: AI Project Data Steward Policy: AI Governance Policy; Project Data Handling Standard (review at each lifecycle gate)	ML Engineer; AI Governance Reviewer (read-only)	ML Engineer (cannot alter project linkage key)	No role holds link rights; linkage run at GOV-LIN-05 under AI Project Data Steward approval	GOV-CLS-04; GOV-STW-04; GOV-ACC-06; GOV-ACC-07; GOV-LIN-05

Lineage Checkpoint and Access Audit Findings

Checkpoint	Input/Source	Transformation/Destination	Audit Finding	Access Finding	Escalation Flag	Corrective Action	Evidence ID(s)
GOV-LIN-01 Imaging ingest	PACS CT study metadata	Approved DICOM metadata extraction → restricted metadata extract	**Complete** - source, transformation version, run ID and access log all recorded	Imaging Data Engineer only, via RBAC + managed identity; RawRadiologySource is read-only; ML Engineer denied	**None**	None required; maintain quarterly steward review	GOV-LIN-01; GOV-ACC-01; GOV-ACC-02; CFG-CONN-01; GOV-STW-01
GOV-LIN-02 Imaging de-identification	Restricted metadata extract	DICOM de-identification profile + project UID replacement → de-identified imaging metadata	**Complete** lineage record. The profile keeps exact StudyDate unchanged (privacy finding, Section 2)	Imaging Data Engineer only; output written to DerivedProjectStore via managed identity	**None** for lineage; cross-reference to the Section 2 date finding	When the project-specific date shift is applied, record the new transformation version and run ID at this checkpoint	GOV-LIN-02; PRV-DEID-01; PRV-RISK-05; CFG-CONN-02
GOV-LIN-03 Report de-identification	Source radiology report text	Automated PHI redaction + QA sampling → redacted report text	**GAP** - source, destination and access log recorded, but transformation version and execution/run ID are blank; fails the completeness rule in CFG-DP-01	Clinical Text Data Engineer only; ML Engineer denied source text; access log linked. No access violation	**ESCALATE** to AI Governance Committee (gate owner) - remediable documentation gap that blocks controlled production	Re-run redaction under a pinned, recorded pipeline version together with the Section 2 safeguard; record version + run ID; repeat QA sample; Clinical Text Data Steward attests the record	GOV-LIN-03; GOV-ACC-03; GOV-ACC-04; GOV-STW-02; CFG-DP-01; CFG-CONN-02; PRV-DEID-02; PRV-POL-03
GOV-LIN-04 Patient attribute preparation	Approved EHR extract	Attribute filtering + age banding → minimum-necessary patient attributes	**Complete** - all checkpoint fields recorded	Clinical Informatics Analyst retrieve/transform/link via role-based access to the project extract	**Advisory** - documentation: 'sex' is listed as a retained attribute but has no de-identification or minimum-necessary entry	EHR Data Steward to confirm whether sex is retained; if so, add a minimum-necessary justification, otherwise correct the dataset record	GOV-LIN-04; GOV-ACC-05; GOV-CLS-03; PRV-DEID-03; PRV-MN-06 to PRV-MN-09
GOV-LIN-05 Multimodal linkage	De-identified imaging metadata + redacted report text + approved patient attributes	Join on project-specific pseudonymous study key → linked de-identified analysis dataset	**Conditional** - checkpoint record complete, but it inherits the GOV-LIN-03 gap, so the redacted-text input cannot be traced end to end	Access event tied to AI Project Data Steward approval; ML Engineer cannot alter linkage key; Governance Reviewer read-only	**ESCALATE** (dependent) - closes only when GOV-LIN-03 is closed; the linked record also carries the combined privacy exposure	Re-run linkage after upstream remediation and record the new run ID; re-check linked data as PRV-RISK-05 requires; confirm the effect on evaluation dataset v1.2	GOV-LIN-05; GOV-ACC-06; GOV-ACC-07; GOV-STW-04; PRV-RISK-04; PRV-RISK-05; CFG-CONN-03

Section 1 narrative analysis

Classification follows the scenario framework. DICOM metadata and report text contain direct identifiers at source, so they are Restricted Clinical Data confined to de-identification roles, and the ML Engineer is explicitly denied source access (GOV-CLS-01, GOV-CLS-02, GOV-ACC-01 to GOV-ACC-04). Structured attributes and the linked dataset carry no direct identifiers but retain linked health information, so they are Confidential Derived Health Data restricted to the project workspace (GOV-CLS-03, GOV-CLS-04). Every asset has a named owner, steward and governing policy, and the linked dataset must be reviewed at each lifecycle gate, including this one (GOV-STW-01 to GOV-STW-04).

The lineage audit traced all three inputs to the linked dataset: four of five checkpoints are complete (GOV-LIN-01, GOV-LIN-02, GOV-LIN-04, GOV-LIN-05). Report de-identification records source, destination and access log but not transformation version or run ID (GOV-LIN-03), and linkage inherits that gap because it consumes the redacted text (GOV-LIN-05). This is escalated to the AI Governance Committee as a remediable blocker (CFG-DP-01). The access audit found no violations: all seven permission decisions

are approved, role-enforced and access-logged (GOV-ACC-01 to GOV-ACC-07). One documentation gap remains: 'sex' is listed as retained but has no de-identification or minimum-necessary entry (GOV-CLS-03, PRV-DEID-03).

Section 2 - Privacy Protection Plan

Privacy position statement: De-identification has removed all direct identifiers, but an unmitigated medium re-identification exposure spanning all three data dimensions means privacy protection is not yet adequate for controlled production (PRV-RISK-04, PRV-POL-05).

Re-identification Risk Assessment

Data Dimension	Exposure/Indicator	De-identification Method	Minimum-Necessary Finding	Risk Level	Safeguard Recommendation	Policy Alignment Flag	Evidence ID(s)
Imaging metadata	Direct identifiers (PatientName, PatientID, AccessionNumber, StudyInstanceUID)	Direct identifiers removed; StudyInstanceUID replaced with project UID. 100-record automated rule check: identifiers absent, UIDs unique. Appropriate for structured DICOM headers	Modality and BodyPartExamined justified (CT cohort routing and abdominal cohort confirmation)	**Low**	No additional safeguard; keep the rule check in the pipeline	**Aligned** - PRV-POL-01	PRV-DEID-01; PRV-MN-01; PRV-MN-02; PRV-POL-01
Imaging metadata	Exact StudyDate retained in all 1,200 studies	Retained unchanged	**Not justified** - only chronological order is needed, and a relative day or project-shifted date preserves sequence	**Low** alone; contributes to the **Medium** combined exposure	Apply a consistent project-specific date shift (or relative interval) and document the rationale	**Misaligned** - PRV-POL-02 trigger met (exact date without documented necessity)	PRV-DEID-01; PRV-MN-03; PRV-RISK-01; PRV-POL-02
Report text	Direct identifiers (names, MRNs, dates, locations, provider names)	Automated PHI redaction + 10% manual QA sample; no direct identifiers found in sample. Appropriate for narrative text, but the pipeline version is not recorded (GOV-LIN-03)	Impression section justified for multimodal context after redaction	**Low**	Re-run under a recorded pipeline version and repeat the QA sample after the safeguard is applied	**Aligned** - PRV-POL-01	PRV-DEID-02; PRV-MN-04; PRV-POL-01; GOV-LIN-03
Report text	Rare phrase 'retired lighthouse keeper from Harbor Point' (1 of 1,200 reports)	Automated redaction did not remove it; detected at manual QA	**Not justified** - no documented model or operational requirement; not needed for the model task	**Medium** - unique occupational and location detail	Redact or generalize the phrase, then repeat the QA sample	**Misaligned** - PRV-POL-03 trigger met (unusually specific contextual detail)	PRV-DEID-02; PRV-MN-05; PRV-RISK-02; PRV-POL-03
Linked patient attributes	Age band 85–89 + rare comorbidity flag (4 of 1,200 linked records)	Source ID removed; exact age converted to age band; address/ZIP excluded; service line, encounter type and approved comorbidity flags retained. Appropriate generalization	Age band, service line and encounter type justified. Rare comorbidity flag only **temporarily** justified for subgroup evaluation, with review due before controlled production. 'Sex' has no entry	**Low** alone (small cell, n = 4); contributes to the **Medium** combined exposure	Confirm subgroup validation is complete, then suppress the rare comorbidity flag; justify or remove 'sex'	**Conditional** - PRV-POL-04 linkage-set assessment; PRV-MN-09 review outstanding	PRV-DEID-03; PRV-MN-06 to PRV-MN-09; PRV-RISK-03; PRV-POL-04; GOV-CLS-03

Data Dimension	Exposure/Indicator	De-identification Method	Minimum-Necessary Finding	Risk Level	Safeguard Recommendation	Policy Alignment Flag	Evidence ID(s)
Intersection of all three dimensions	Exact StudyDate + unusual phrase + 85–89 age band / rare comorbidity in the same record	No direct identifiers remain, but no cross-dimension safeguard has been applied	Two of the three contributing elements lack justification (PRV-MN-03, PRV-MN-05)	**Medium**	Implement the proposed safeguard (date shift + phrase removal/generalization), re-check the linked data, document residual risk, obtain Privacy Officer sign-off	**Not aligned** at gate - PRV-POL-05: unmitigated medium risk blocks controlled production	PRV-RISK-04; PRV-RISK-05; PRV-POL-04; PRV-POL-05; CFG-DP-02

Section 2 narrative analysis

Each method suits its data type (PRV-DEID-01 to PRV-DEID-03). Imaging metadata had identifiers removed and UIDs replaced, confirmed by an automated rule check (PRV-DEID-01); report text received automated PHI redaction with a 10% manual QA sample (PRV-DEID-02); and patient attributes were generalized to age bands with address/ZIP excluded (PRV-DEID-03). The weaknesses lie in what was retained. Exact StudyDate fails minimum-necessary review because a shifted or relative date preserves order (PRV-MN-03, PRV-POL-02). The phrase 'retired lighthouse keeper from Harbor Point' has no documented purpose (PRV-MN-05, PRV-POL-03), and the rare comorbidity flag is justified only until subgroup review (PRV-MN-09).

Individually modest, these exposures converge in one record: an exact date, a phrase unique among 1,200 reports, and the 85–89 age band with the rare comorbidity, a combination shared by only four records (PRV-RISK-01 to PRV-RISK-04). Policy requires an applied safeguard and privacy sign-off for medium risk, yet the proposed date shift and phrase generalization have no implementation evidence (PRV-POL-05, PRV-RISK-05).

These findings connect directly to Section 1. The residual phrase passed through the same report de-identification step that lacks a version and run ID, so the redaction configuration that missed it cannot be identified (GOV-LIN-03, PRV-DEID-02). One coordinated remediation should apply the date shift, generalize the phrase under a recorded pipeline version, repeat QA sampling and re-run linkage, closing the lineage gap and producing the evidence the Privacy Officer needs (CFG-DP-01, CFG-DP-02). The residual risk also confirms the Confidential classification of the linked dataset (GOV-CLS-04).

Section 3 - Project Configuration and Operational Readiness Assessment

Configuration/readiness position statement: Workspace, model access and evaluation flows are ready, but data connections carry the open report-text lineage gap and three of five transition criteria remain unmet, so the project is not ready for controlled production (CFG-CONN-02, CFG-DP-01 to CFG-DP-06).

Configuration Review Findings

Course-Defined Review Area	Evidence Reviewed	Finding	Readiness Status	Corrective Action/Recommendation	Evidence ID(s)
Review Area 1 - Project / Workspace Configuration	Access model, network access, audit logging, environment separation, controlled-production environment	Least-privilege role groups (GovernanceReviewer read-only; MLEngineer contributor; DataEngineer connection operator); public network disabled; access and deployment actions logged; development and test separated by resource group and access group; production access group exists with no active deployment	**Ready**	Keep the controlled-production environment inactive until the CFG-DP-06 decision; supply audit-log extracts as gate evidence	CFG-WS-01 to CFG-WS-05
Review Area 2 - Model Access	Approved model, deployment permission, reviewer access, version lock	ImagingTriageModel-v3 is the governance-approved version (record AI-GOV-27); deployment restricted to ML Platform Engineer; AI Governance Reviewer read-only (consistent with GOV-ACC-07); v3 pinned and evaluation linked to the v3 hash	**Ready**	The production recommendation must cite the v3 hash; re-confirm the v3 evaluation if remediation changes evaluation data	CFG-MOD-01 to CFG-MOD-04; GOV-ACC-07
Review Area 3 - Data Connections	RawRadiologySource, DerivedProjectStore, EvaluationStore, secrets handling	All connections use managed identity and no connection strings are embedded; the source connection is read-only. DerivedProjectStore relies on the incomplete GOV-LIN-03 record and holds data whose privacy safeguard has not been applied	**Not ready** (remediable)	Close GOV-LIN-03, apply the safeguard in DerivedProjectStore and refresh lineage references. Clarify how the ML Engineer 'Transform' right on the linked dataset maps to read-only connections	CFG-CONN-01 to CFG-CONN-04; GOV-LIN-03; GOV-ACC-06; PRV-RISK-05
Review Area 4 - Evaluation Flows	Locked dataset, sensitivity, PPV, sign-off	Locked dataset v1.2 (300 studies, hash recorded); sensitivity 0.91 vs ≥ 0.90; PPV 0.76 vs ≥ 0.75; clinical informatics and AI governance sign-offs recorded. Each metric clears its threshold by only 0.01	**Ready**	If remediation alters records in v1.2, lock a new version and re-confirm thresholds. Monitoring alerts do not cover sensitivity or PPV (CFG-MON-04); given the thin margins, add them	CFG-EVAL-01 to CFG-EVAL-04; CFG-MOD-04; CFG-MON-04

Transition Criteria Assessment

Transition Criterion	Evidence	Assessment	Required Action	Evidence ID(s)
CFG-DP-01 - Complete lineage for every production-bound input (owner: AI Governance Committee)	GOV-LIN-01, -02, -04 and -05 complete; GOV-LIN-03 has blank transformation version and run ID	**NOT MET** (remediable)	Re-run report redaction with recorded version and run ID; re-run GOV-LIN-05 linkage; committee verifies all five checkpoints	CFG-DP-01; GOV-LIN-01 to GOV-LIN-05
CFG-DP-02 - No unresolved medium/high re-identification exposure without approved safeguard (owner: Privacy Officer)	Medium combined exposure identified; safeguard proposed but not applied; privacy sign-off pending	**NOT MET**	Apply date shift and phrase removal/generalization, re-check linked data, attach implementation evidence, obtain privacy sign-off	CFG-DP-02; PRV-RISK-04; PRV-RISK-05; PRV-POL-05
CFG-DP-03 - Evaluation thresholds met on locked dataset (owner: Clinical Informatics + AI Governance)	Locked v1.2; sensitivity 0.91 (≥ 0.90); PPV 0.76 (≥ 0.75); both sign-offs recorded	**MET**	No action unless privacy remediation changes v1.2 contents; then re-confirm on the new locked version	CFG-DP-03; CFG-EVAL-01 to CFG-EVAL-04
CFG-DP-04 - Capacity at 100% expected volume with p95 ≤ 5.0 s and throttling < 1.0% (owner: ML Platform Lead)	Tested only to 75% (900 studies/day): p95 4.8 s, throttling 0.8%, utilization 86%; full-volume validation not run	**NOT MET** (remediable)	Run the full-volume test at 1,200 studies/day; scale capacity if targets are missed; record cost at the passing configuration	CFG-DP-04; CFG-MON-01 to CFG-MON-03

Transition Criterion	Evidence	Assessment	Required Action	Evidence ID(s)
CFG-DP-05 - Monitoring, alerting and rollback controls configured (owner: ML Platform Lead)	Dashboard and alerts for p95 latency, throttling, utilization, error rate and daily cost; rollback runbook names prior model version, traffic-disable procedure, owner and escalation contacts	**MET**	None blocking; exercise the rollback runbook during the full-volume test	CFG-DP-05; CFG-MON-04; CFG-MON-05

Monitoring and Cost-Performance Assessment

Metric/Signal	Observed Evidence	Scenario Target	Assessment	Recommendation	Evidence ID(s)
p95 inference latency	3.2 s at 600/day (50%) → 4.8 s at 900/day (75%); not measured at 100%	≤ 5.0 s	**At limit** - within target but 0.2 s headroom at 75%. Rose 1.6 s per 25-point volume step; linear projection about 6.4 s at 100% (projection, not measured evidence)	Run the full-volume test; add inference capacity before controlled production if p95 exceeds 5.0 s	CFG-MON-01; CFG-MON-02; CFG-MON-03
Throttled requests	0% at 50% → 0.8% at 75%	< 1.0%	**At limit** - within target but near the ceiling; linear projection about 1.6% at 100% would breach the limit	Raise throughput capacity or quota and confirm < 1.0% at full volume; keep the throttling alert active	CFG-MON-01; CFG-MON-02; CFG-MON-04
Peak resource utilization	58% at 50% → 86% at 75%	No numeric limit; utilization alert required	**Headroom** signal - +28 points per 25-point step leaves too little room for the remaining 25% of volume; configured resources are unlikely to support 1,200 studies/day	Size capacity for 100% volume plus operating headroom and validate it in the full-volume test	CFG-MON-01; CFG-MON-02; CFG-MON-04; CFG-DP-04
Estimated daily cost	USD 43 at 600/day → USD 61 at 900/day (about USD 0.06 per additional study)	≤ USD 80/day	**Within** budget at current capacity - linear projection about USD 79/day at 1,200, leaving about USD 1 of headroom. Extra capacity to fix latency or throttling will likely exceed budget	Record cost in the full-volume test at the configuration that meets performance targets; if above USD 80/day, escalate the budget decision to the AI Governance Committee rather than accept degraded performance	CFG-MON-01; CFG-MON-02; CFG-MON-03; CFG-DP-06

Deployment pathway recommendation: Do not progress to controlled production; remain in the testing state, because CFG-DP-06 permits progression only when every criterion is evidenced as met. Evaluation (CFG-DP-03) and monitoring/rollback (CFG-DP-05) are met, but lineage completeness (CFG-DP-01), privacy safeguard and sign-off (CFG-DP-02) and full-volume capacity validation (CFG-DP-04) are not. All three gaps are remediable; resubmit to the AI Governance Committee once closure evidence is attached (CFG-DP-01, CFG-DP-04, PRV-POL-05).

Section 3 narrative analysis

The four review areas and the testing and controlled-production states are course- and scenario-defined, not native Microsoft Foundry stages (CFG-DP-06). Workspace controls are sound: role-separated groups, private networking, audit logging, separated environments and an inactive production environment (CFG-WS-01 to CFG-WS-05). Model access is limited to the approved, pinned ImagingTriageModel-v3 (CFG-MOD-01 to CFG-MOD-04). Evaluation passed with dual sign-off, though sensitivity and PPV each clear their thresholds by only 0.01 (CFG-EVAL-01 to CFG-EVAL-04). Connections use managed identities without embedded secrets, but DerivedProjectStore depends on the incomplete GOV-LIN-03 record (CFG-CONN-02, CFG-CONN-04).

Monitoring reveals a cost-performance tradeoff: from 50% to 75% of expected volume, p95 latency rose from 3.2 to 4.8 seconds, throttling from 0% to 0.8% and peak utilization from 58% to 86% (CFG-MON-01, CFG-MON-02). Linear projection to 1,200 studies/day gives about 6.4 seconds and 1.6% throttling, both outside target, so current resources are unlikely to sustain expected volume; only the unrun full-volume test can confirm this (CFG-MON-03). Cost projects to about USD 79 against a USD 80 budget, so the capacity needed to hold latency will probably exceed budget and requires an explicit decision (CFG-MON-02, CFG-DP-04).

Configuration also depends on Sections 1 and 2. Lineage and privacy remediation will change DerivedProjectStore and the linkage output feeding EvaluationStore (CFG-CONN-02, CFG-CONN-03, GOV-LIN-05). If affected records sit in dataset v1.2, its hash changes and the v3 evaluation must be re-confirmed (CFG-MOD-04). The sequence should be privacy and lineage remediation, then evaluation re-confirmation, then the full-volume load test (CFG-DP-02 to CFG-DP-04).

Final Cross-Section Check

[x] Every finding, escalation flag, and recommendation cites at least one Evidence ID.

[x] Cross-section dependencies are explicitly identified (Section 2, final paragraph; Section 3, final paragraph; Tables 2-5).

[x] All required supporting tables are complete (rows matched to the evidence: 4 datasets, 5 lineage checkpoints, 6 exposures, 4 review areas, 5 transition criteria, 4 monitoring signals).

[x] The narrative remains within 700-900 words (849 words across the position statements, narrative analyses and deployment pathway recommendation; tables excluded).